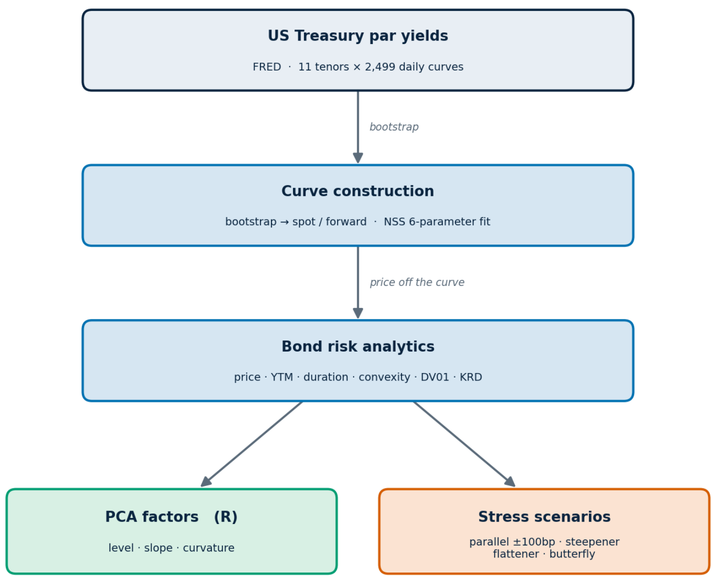
Rates Risk Engine

US Treasury Curve Construction & Fixed-Income Risk Analytics



Bootstraps the US Treasury spot curve from constant-maturity par yields, calibrates a Nelson-Siegel-Svensson model, prices a ten-bond portfolio off the curve, measures interest rate risk through DV01 and key rate durations, decomposes historical curve moves into level, slope, and curvature factors, and runs parallel and non-parallel stress scenarios.

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Valuation date: 2025-12-31

1. Data and Bootstrap

Daily US Treasury constant-maturity par yields for eleven tenors (1M to 30Y) are sourced from FRED. Par yields are interpolated onto a semiannual grid with a monotone cubic spline, and discount factors are recovered by forward substitution on the par-bond identity $1 = (c / 2) \cdot \text{sum}\ D(t_i) + D(t_n)$. Continuously compounded spot rates follow from $z_n = -\ln D(t_n) / t_n$ and interval forward rates from successive discount factors.

The bootstrap is validated by repricing the par bonds off the recovered discount factors: each returns to par to within numerical tolerance, confirming internal consistency of the spot curve.

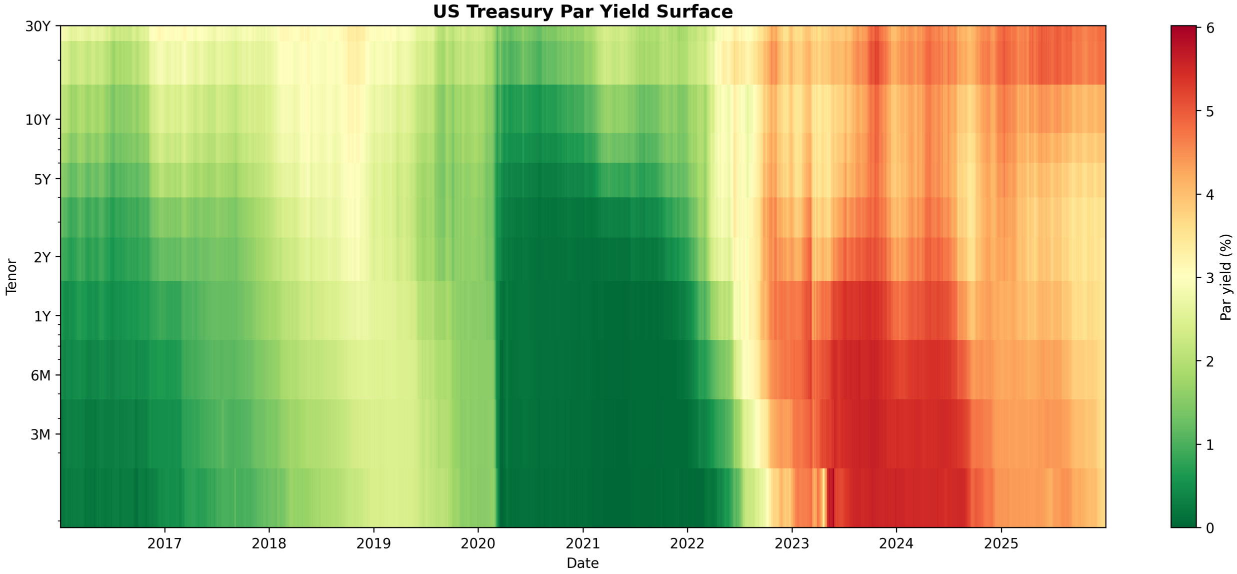

Figure 1. Par yield surface over the sample. The red band marks the 2022-2024 curve inversion, where short tenors yield more than long tenors.

2. Nelson-Siegel-Svensson Calibration

The spot curve is fitted with the six-parameter Nelson-Siegel-Svensson form by nonlinear least squares. The coefficients carry economic meaning: *b0* is the long-run level, *b1* the short-end slope, and *b2, b3* two curvature (hump) terms with decay parameters *tau1, tau2*. Across 2,499 fitted curves the median fit error is 1.74 bp and the maximum is 12.16 bp.

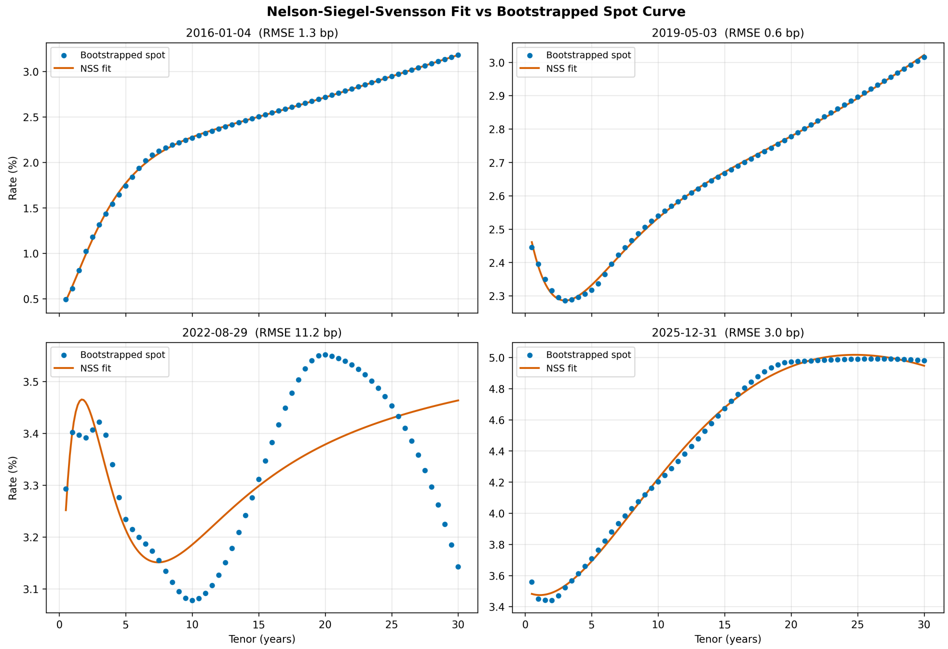

Figure 2. NSS fitted curves (orange) against bootstrapped spot rates (blue) on four representative dates spanning the sample.

3. Bond Risk Analytics

Each bond is priced by discounting its cash flows off the spot curve on 2025-12-31. Yield to maturity is the single continuous rate that reproduces the price; Macaulay duration is the present-value weighted average cash flow time, and convexity the second moment. DV01 is the price change for a one-basis-point parallel shift.

Pricing and sensitivities

Bond	Price	YTM	Mod Dur	Convexity	DV01
CORP_10Y	1,128,850	4.12%	7.90	72.0	891
CORP_5Y	1,080,318	3.69%	4.46	21.4	482
UST_10Y	2,011,490	4.14%	8.26	76.8	1,661
UST_10Y_STRIP	656,946	4.20%	10.00	100.0	657
UST_20Y	961,853	4.74%	13.25	225.3	1,274
UST_2Y	1,014,944	3.44%	1.94	3.8	197
UST_30Y	1,891,041	4.79%	16.36	380.9	3,094
UST_3Y	1,012,709	3.52%	2.86	8.4	289
UST_5Y	1,012,252	3.69%	4.58	22.2	464
UST_7Y	1,009,776	3.90%	6.16	41.1	622

The zero-coupon bond (UST_10Y_STRIP) has Macaulay duration equal to its ten year maturity, the analytic benchmark that confirms the duration implementation.

Key rate durations

Key rate durations isolate exposure to each segment of the curve by applying localised tent-shaped shifts at the 2Y, 5Y, 10Y, 20Y, and 30Y nodes. Because the tent functions sum to a parallel shift, the row sum of the key rate durations reconciles to the parallel DV01.

Bond	KRD 2Y	KRD 5Y	KRD 10Y	KRD 20Y	KRD 30Y	Sum
CORP_10Y	25.6	103.0	750.9	0.0	0.0	879.5
CORP_5Y	24.5	449.6	0.0	0.0	0.0	474.1
UST_10Y	37.9	152.2	1,452.6	0.0	0.0	1,642.7
UST_10Y_STRIP	0.0	0.0	656.6	0.0	0.0	656.6
UST_20Y	20.0	80.6	227.7	913.7	0.0	1,242.0
UST_2Y	190.7	0.0	0.0	0.0	0.0	190.7
UST_30Y	40.1	161.2	455.3	659.0	1,675.8	2,991.4
UST_3Y	191.1	92.5	0.0	0.0	0.0	283.6
UST_5Y	17.8	440.2	0.0	0.0	0.0	458.0
UST_7Y	18.3	374.2	222.9	0.0	0.0	615.4

4. Principal Component Analysis

Daily changes in the par curve are decomposed by PCA. The first three components reproduce the canonical level, slope, and curvature factors and together explain the overwhelming majority of curve variance, the empirical basis for the parallel and non-parallel stress scenarios in the next section.

Component	Eigenvalue	Explained	Cumulative
PC1	1.915e-06	79.9%	79.9%
PC2	2.709e-07	11.3%	91.2%
PC3	1.100e-07	4.6%	95.8%

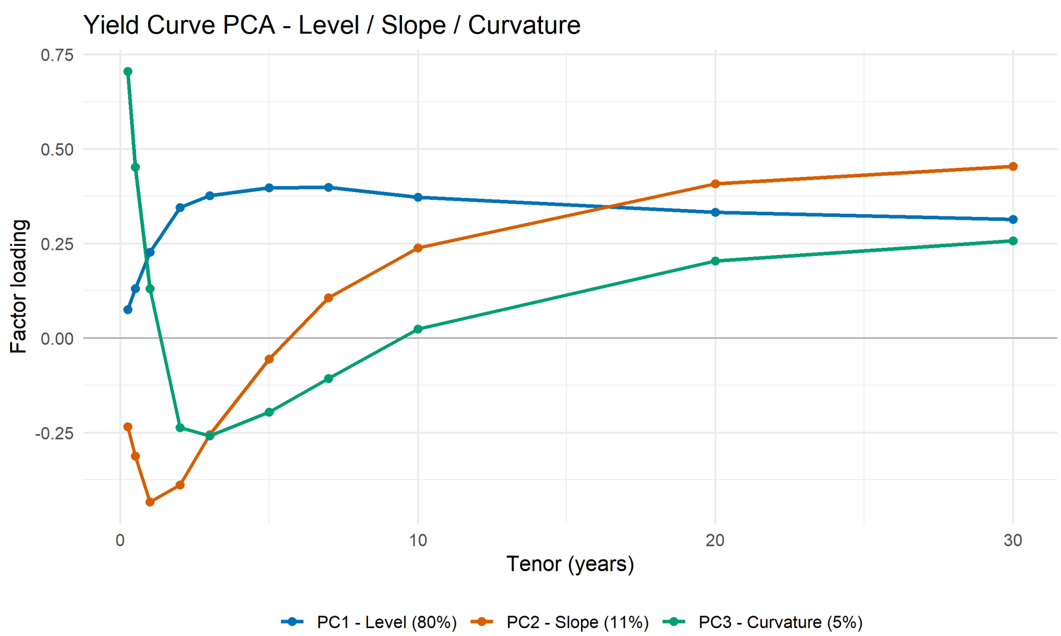

Figure 3. Loadings of the first three principal components. PC1 is flat (level), PC2 is monotone in tenor (slope), and PC3 is convex (curvature).

5. Stress Testing

The portfolio is fully repriced under parallel, steepener, flattener, and butterfly shocks. Full revaluation captures convexity, so for a parallel shift the realised loss is slightly smaller than the linear DV01 estimate; the difference is the convexity contribution.

Scenario	Portfolio P&L
butterfly	$-182,402
flattener	$321,218
parallel_down_100	$1,020,598
parallel_up_100	$-890,109
steepener	$-292,641

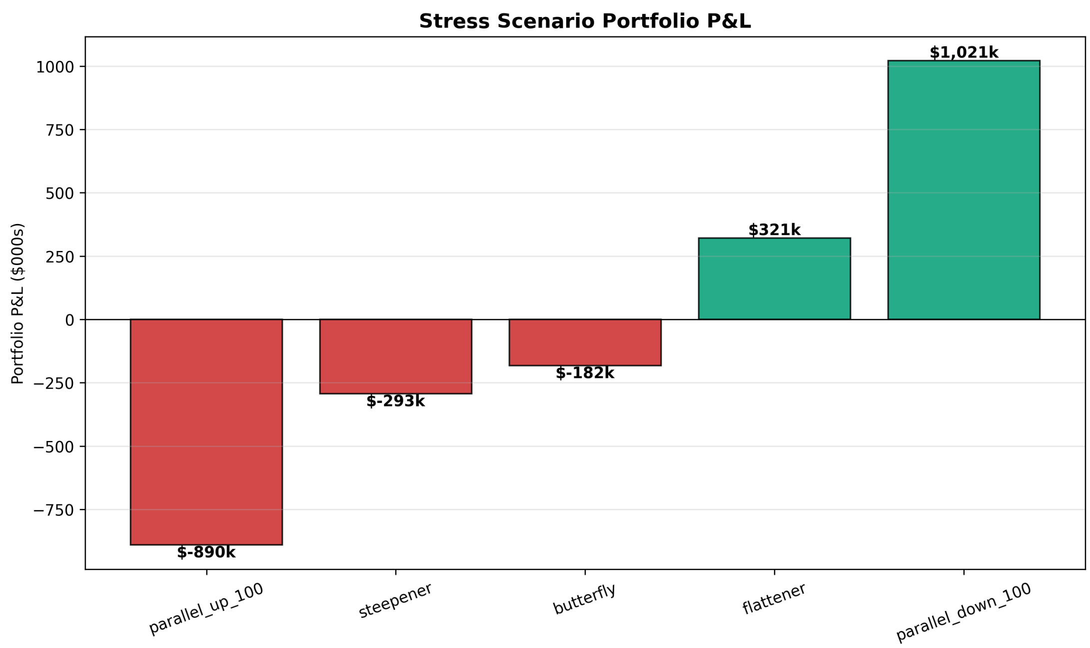

Figure 4. Portfolio profit and loss under each stress scenario.

References

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